CSE 11 Lecture 10

Answer Box #1

What were 3 custom filters others around you wrote?

How do you think Scan worked if you had AI generate it? Same as those around you? Are you confident it works?

Color(r, g, b, a)

a: 0 - 255

Property-based test: inputs where you can easily compute/check expected output

(like digits → image → digits

check equality

Reading Pict.java, Again

Name: _____ Date: _____

Some of `Pict.java`, exerpted verbatim from the source file.

```
final class Px {
  private Px() {}
  static int of(double v) {
    if (Double.isNaN(v)
        || Double.isInfinite(v)) {
      throw new IllegalArgumentException(
        "measurement must be a number, got "
        + v);
    }
    return (int) Math.round(v);
  }
  static int size(String what, int v) {
    if (v < 0) {
      throw new IllegalArgumentException(
        what + " is negative: " + v);
    }
    return v;
  }
}
```

```
record Circle(int radius, Color color)
    implements Pict {
  Circle { Px.size("radius", radius); }
  Circle(int radius, String color) {
    this(radius, Colors.named(color));
  }
  Circle(double radius, String color) {
    this(Px.of(radius), Colors.named(color));
  }
  // ...
}
// Rect has the same three constructors,
// with a width and a height.
```

runs extra checks before the record constructor

v = 10.5

round ret "long"

→ Color.RED

int: $-2^{31} \leftrightarrow 2^{31}-1$

long: $-2^{63} \leftrightarrow 2^{63}-1$

record Rect(int width, int height, Color color){

Rect(int w, int h, String color)

Rect(double w, double h, String color)

Which constructor?

Q1. For each line, write the parameter types of the constructor Java runs *first*, and what prints.

you wrote	runs first	prints
`new Circle(10, "red")`	int, String	Circle[radius=10, color=Color[r=255 ...]
`new Circle(10.5, "red")`	double, String	Circle[radius=11]
`new Rect(3, 4.5, "red")`	double, double, String	Rect[width=3, height=4.5]

3.0 gets passed in

A crash, three calls deep

Running `Pict p = new Circle(-0.6, "red");` in `main` crashes with:

```
Exception in thread "main" java.lang.IllegalArgumentException: radius is negative: -1
  at Px.size(Pict.java:90)
  at Circle.<init>(Pict.java:143)
  at Circle.<init>(Pict.java:145)
  at Main.main(Main.java:3)
```

(`<init>` is how a stack trace names a constructor.)

Q2. Draw the stack at the moment `Px.size` throws. Write each call as it appears in the code, and the values of its parameters.

NAMES (THE STACK) HEAP

Answer Box #2

_____
what = _____ v = _____

Circle(int, Color)
radius = -1 color = Color.RED

(this code calls Px.size)

Circle(double, String)
radius = -0.6 color = "red"

main()
p = *not set yet*

top level

Q3. The radius written was `-0.6`, but the message says `-1`. Which line turned one into the other?

What Px.of gives back

Q4. What does each call return? If it throws, write the message.

`Px.of(2.5)`	
`Px.of(-2.5)`	
`Px.of(2.49)`	
`Px.of(Double.NaN)`	
`Px.of(1e10)`	

Q5. `(int) 1e10` would be `2147483647`, the biggest `int`. But `Px.of(1e10)` is `1410065408`. What's different about the way `Px.of` converts?

Who gets called on what

Q6. Is each one called on an object, or on a class? Circle one.

`Px.of(2.5)` object / class

`pic.width()` object / class

`Math.round(2.5)` object / class

`Colors.named("red")` object / class

`r.colorAt(0, 0)` object / class

`"trash".length()` object / class

Raster

```
class Raster {
  private final Color[][] pixels;
  private final int width, height;
  // ... constructors that check and copy the pixels
  int width() { return this.width; }
  Color colorAt(int x, int y) { return this.pixels[y][x]; }
  public boolean equals(Object other) {
    // ... same size, and every row Arrays.equals the other's
  }
}
```

Q7. In your own program, `r` is a `Raster`. Which lines compile? Circle one.

`int w = r.width();` compiles / doesn't

`int w = r.width;` compiles / doesn't

`Color c = r.colorAt(0, 0);` compiles / doesn't

`Color[][] all = r.pixels;` compiles / doesn't

Q8. `r1` and `r2` are two `Raster`s made from the same colors. What are `r1.equals(r2)` and `r1 == r2`?

`r1.equals(r2)` _____ `r1 == r2` _____

Q9. Suppose `Raster` were a record, `record Raster(Color[][] pixels, int width, int height)`. Give two things that could go wrong.